

#26 Competitor Price Scraper and Alert

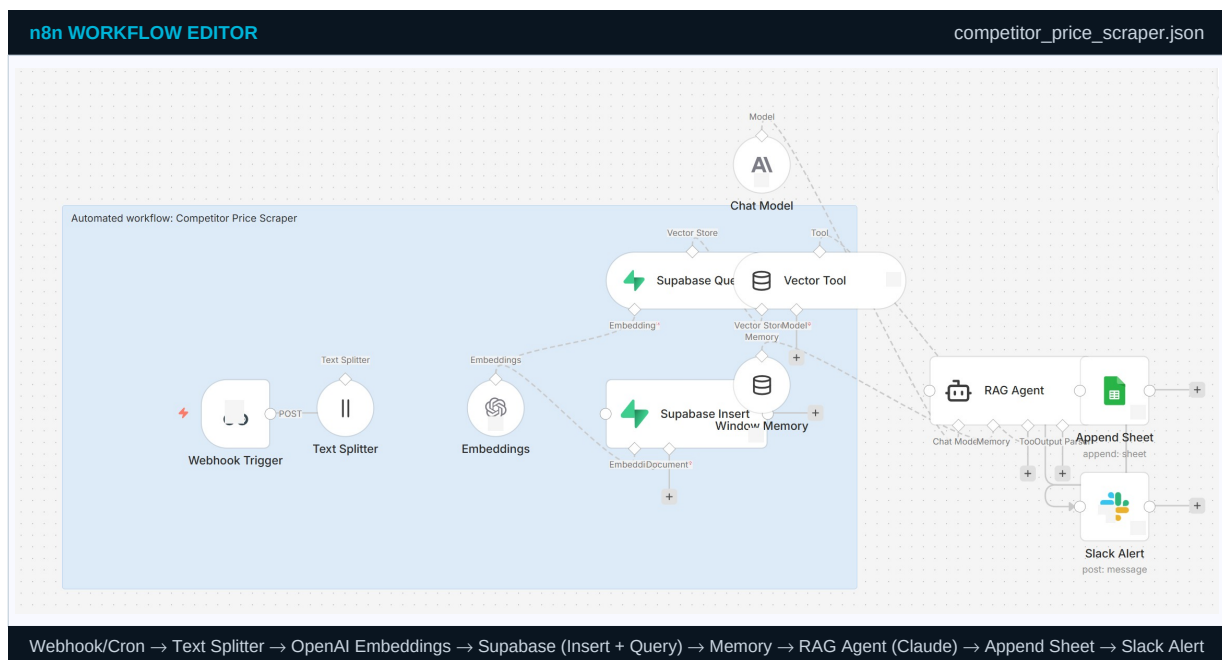
Data Analytics /
Competitive
Intelligence

Know the moment a competitor changes price — before your team does

BUSINESS PROBLEM

Pricing teams check competitor sites once a week. This workflow scrapes competitor pricing pages daily at 06:00, detects any change using vector semantic diff rather than fragile HTML selectors, and delivers a structured change brief via Slack within minutes of detection.

n8n CANVAS — actual workflow screenshot



INPUT / OUTPUT — exact n8n JSON format

INPUT — Cron trigger 06:00 daily — competitor URL config

```
{
  "run_date": "2024-06-15",
  "competitors": [
    { "name": "Competitor A", "url": "https://competitora.com/pricing" },
    { "name": "Competitor B", "url": "https://competitorb.com/plans" }
  ],
  "change_threshold": 0.15
}
```

OUTPUT — Slack alert — detected price change with before/after and strategy note

```
{
  "json": {
    "competitor": "Competitor A",
    "change_type": "PRICE_INCREASE",
    "plan_name": "Business 500Mbps",
    "before": "$79/month",
    "after": "$89/month",
    "pct_change": 12.66,
    "change_added": "Free installation (was $99)",
    "semantic_diff": 0.82,
    "strategic_read": "Opportunity to offer $84 competitive rate",
    "detected_at": "2024-06-15T06:14:22Z"
  }
}
```

STEP-BY-STEP — what each node does

#	WHAT HAPPENS	PRACTICAL DETAIL
1	■ Cron at 06:00 — reads competitor URL list	Daily trigger reads competitor URL config from Google Sheets. Processes each URL sequentially.
2	✂️ Text Splitter — page content chunks	Pricing page content chunked into 300-char segments for granular semantic change detection.
3	■ OpenAI Embeddings — content vectors	Price tables, plan names, and feature lists embedded separately for precise change detection.
4	■ Supabase Insert — store today's snapshot	Stores: {competitor, page_url, chunk_text, embedding, snapshot_date} for every chunk.
5	■ Supabase Query — yesterday's snapshot	Fetches previous day's vectors for the same competitor and page for semantic comparison.
6	■ Vector Tool — semantic diff	Agent compares old vs new vectors. High cosine distance signals a content change.
7	■ Window Memory — weekly change history	Tracks 7 days. Detects patterns: 'Competitor A changed pricing 3 times this month.'
8	■ Claude — change analysis	Classifies change type, extracts before and after values, generates strategic interpretation.
9	■ RAG Agent — formats alert	{competitor, change_type, before, after, pct_change, additions, strategic_read, urgency}.
10	■ Append Sheet — competitive log	All changes logged. Powers competitive pricing dashboard in Looker Studio.
11	■ Slack Alert — immediate notification	Posted to #competitive-intel within 15 minutes. Includes before/after comparison table.

CREDENTIALS REQUIRED

CREDENTIAL	WHERE TO GET IT	n8n TYPE
OpenAI API Key	platform.openai.com	OpenAI API
Supabase URL + Anon Key	app.supabase.com → Settings → API	Supabase API
Anthropic (Claude) API Key	console.anthropic.com → API Keys	Anthropic API
Slack Bot Token	api.slack.com/apps → OAuth & Permissions	Slack API

SAMPLE RUN — what this workflow processed and produced

Scenario	A managed connectivity provider monitors 4 competitor pricing pages daily. Previously missed 3 competitor price changes and only discovered them when losing deals.
Input received	06:00 cron. 4 pages scraped. Supabase stored 47 new chunks. Previous day's 44 chunks retrieved for comparison.
What the workflow found	Competitor A's Business 500Mbps plan section: cosine distance 0.82 — well above 0.15 threshold. Claude identified: price \$79 → \$89, free installation added.

Output produced	Slack alert at 06:14. Sales director authorised matched pricing at \$84 by 07:02. Three active quotes updated before business day started. Won 2 of 3 deals citing competitive response.		
JSON: competitor_price_scraper.json	GitHub: trm-competitor-price-monitor	www.similarweb.com/	